PROJECT STRUCTURE

Project Name: Practical_1

- **Source Packages**
  - **practical_1**
    - **Practical_1.java**
      → Contains Java code to establish database connectivity using JDBC.
- **Libraries**
  - **mysql-connector-j-9.5.0.jar**
    → MySQL JDBC Driver required to connect Java application with MySQL database.
  - **JDK 23 (Default)**
    → Java Development Kit used to compile and run the program.

PRACTICAL 1

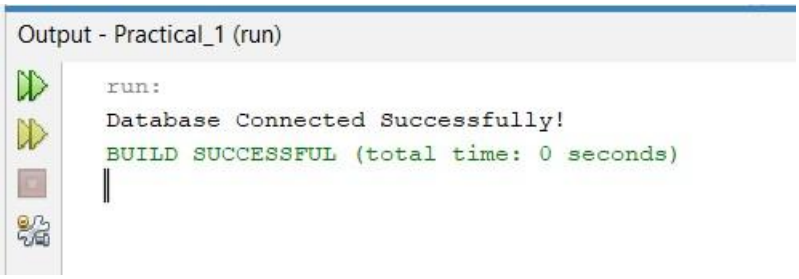
Write a Java program using JDBC to check database connectivity and display an appropriate success or failure message.

CODE:

```
import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;


public class Practical_1 {

    public static void main(String[] args) {

        Connection con = null;

        try {

            // Load JDBC Driver

            Class.forName("com.mysql.cj.jdbc.Driver");


            // Database URL

            String url = "jdbc:mysql://localhost:3307/college";

            String user = "root";

            String password = "";   // XAMPP default


            // Create Connection

            con = DriverManager.getConnection(url, user, password);

            System.out.println("Database Connected Successfully!");

        } catch (ClassNotFoundException e) {

            System.out.println("Error: JDBC Driver not found");

        } catch (SQLException e) {

            System.out.println("Error: Database connection failed");

            System.out.println("Reason: " + e.getMessage());

        } finally {

            try {
```

```
            if (con != null) {
                con.close();
            }
        } catch (SQLException e) {
            System.out.println("Error while closing connection");
        }
    }
  }
}
```

OUTPUT:



PRACTICAL 2

JDBC Connectivity & CRUD Operations Develop a Menu driven java application for student information which will create the table with appropriate columns. Menu will contain the options like insert, update and delete. Based on the option, data will be inserted or updated or deleted from table based on student id (student_id will be primary key). Display appropriate message for each operation.

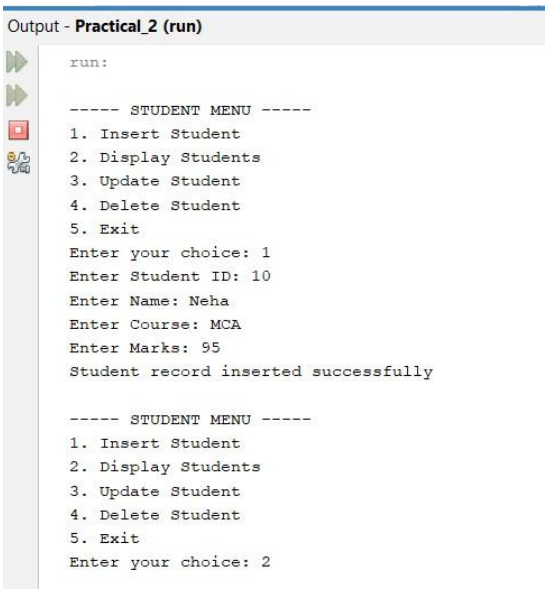
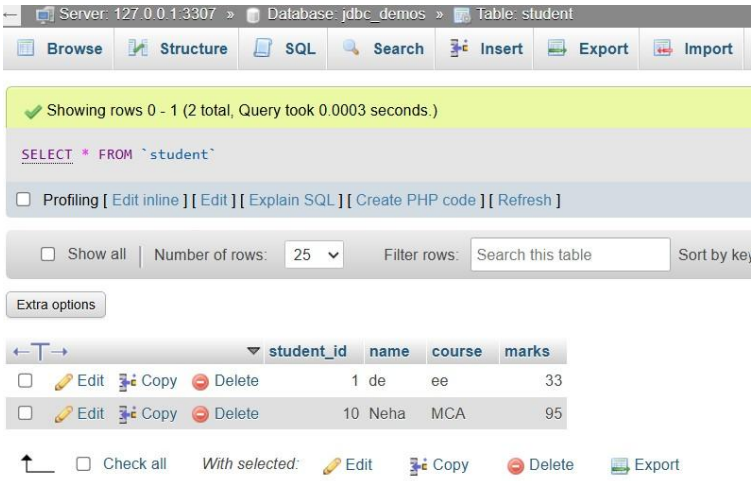
CODE:

```
package practical_2;

import java.sql.*;

import java.util.Scanner;


public class Practical_2 {

    public static void main(String[] args) {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con =
DriverManager.getConnection("jdbc:mysql://localhost:3307/jdbc_demo","root","");


            Statement stmt = con.createStatement();

            Scanner sc = new Scanner(System.in);

            String createTable =

                "CREATE TABLE IF NOT EXISTS Stud (" +

                "student_id INT PRIMARY KEY, " +

                "name VARCHAR(50), " +

                "course VARCHAR(50), " +

                "marks INT)";

            stmt.executeUpdate(createTable);

            int choice;
```

```
do {
    System.out.println("\n----- STUDENT MENU -----");
    System.out.println("1. Insert Student");
    System.out.println("2. Display Students");
    System.out.println("3. Update Student");
    System.out.println("4. Delete Student");
    System.out.println("5. Exit");
    System.out.print("Enter your choice: ");
    choice = sc.nextInt();
    sc.nextLine();
    switch (choice) {
        // INSERT
        case 1:
            System.out.print("Enter Student ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Course: ");
            String course = sc.nextLine();
            System.out.print("Enter Marks: ");
            int marks = sc.nextInt();
            sc.nextLine();
            String insert =
                "INSERT INTO student VALUES (" +
                id + ",'" + name + "','" + course + "'," + marks + ")";

            stmt.executeUpdate(insert);
            System.out.println("Student record inserted successfully");
```

```
            break;
        // SELECT
        case 2:
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");
            System.out.println("\nStudent Records:");
            while (rs.next()) {
                System.out.println(
                    rs.getInt("student_id") + " " +
                    rs.getString("name") + " " +
                    rs.getString("course") + " " +
                    rs.getInt("marks")
                );
            }
            break;
        // UPDATE
        case 3:
            System.out.print("Enter Student ID to update: ");
            int uid = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter New Course: ");
            String newCourse = sc.nextLine();
            System.out.print("Enter New Marks: ");
            int newMarks = sc.nextInt();
            sc.nextLine();
            String update =
                "UPDATE student SET course='" + newCourse +
                "', marks=" + newMarks +
                " WHERE student_id=" + uid;
            int rowsUpdated = stmt.executeUpdate(update);
```

```
                if (rowsUpdated > 0)
                    System.out.println("Student record updated successfully");
                else
                    System.out.println("Student ID not found");
                break;

            // DELETE
            case 4:
                System.out.print("Enter Student ID to delete: ");
                int did = sc.nextInt();
                sc.nextLine();
                String delete =
                    "DELETE FROM student WHERE student_id=" + did;
                int rowsDeleted = stmt.executeUpdate(delete);
                if (rowsDeleted > 0)
                    System.out.println("Student record deleted successfully");
                else
                    System.out.println("Student ID not found");
                break;

            case 5:
                System.out.println("Exiting program...");
                break;

            default:
                System.out.println("Invalid choice");
        }
    } while (choice != 5);
} catch (ClassNotFoundException e) {
```

```
            System.out.println("JDBC Driver not found");


        } catch (SQLException e) {
            System.out.println("SQL Error: " + e.getMessage());
        }
    }
}
```

OUTPUT: